Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/17/2021
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Project Title:	The Role of P21/CDKN1a Pathway in Microgravity-Induced Bone Tissue Regenerative Arrest - A Spaceflight Study of Transgenic P21/CDKN1a Null Mice in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
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Space Biology Special Category:	None		
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Key Personnel Changes/Previous PI:	November 2021 Report: Dr. Ruth Globus, Co-Investigator in this grant, retired from NASA service (Feb 2020) and although still a Co-I significantly reduced her participation in this work starting in 2020. Dr. Elizabeth Blaber moved from NASA Ames and is currently a faculty member at Rensselaer Polytechnic Institute, although she continues to actively participate in grant activities. Given the Globus and Blaber departures from NASA Ames, Dr. Cassandra Juran was added to the grant as Co-Investigator, and has conducted with Dr. Almeida the bulk of the laboratory work in 2020-2021 associated with preparations for the spaceflight experiment, flight-associated ground control animal experiments, and flight and ground control tissue sample processing and analyses.		
COI Name (Institution):	Blaber, Elizabeth A Ph.D. (Rensselaer Polytechnic Institute) Globus, Ruth Ph.D. (NASA Ames Research Center, Ret.) Shirazi-Fard, Yasaman Ph.D. (NASA Ames Research Center) Juran, Cassandra M Ph.D. (Blue Marble Space Institute of Science)		

Grant/Contract No.:	Internal Project
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Task Description:	Tissue degeneration and the failure of normal tissue regenerative processes in microgravity are increasing concerns for sustaining life in space as the duration of future flight missions increases. Specifically, our laboratory has identified normal adult stem cell-based tissue regenerative processes, such as the formation of new bone, cartilage, immune cells, and blood from bone marrow hematopoietic stem cell precursors, as being particularly sensitive to the stresses of spaceflight in microgravity. The cellular mechanisms activated in mouse bone during spaceflight are not completely understood, but they include induction of oxidative stress and the inhibition of differentiation of marrow mesenchymal stem cells, via activation of CDKN1a/p21-mediated cell cycle arrest in proliferative osteoprecursor cells on the bone surface. The identification of CDKN1a/p21 mRNA and protein as being highly up-regulated by spaceflight in mice is a seminal finding, because this gene is a master regulator of tissue regeneration, that when knocked out allows for mammalian limb regeneration similar to that found in amphibians. Because we established in multiple spaceflight and ground experiments that the CDKN1a/p21 pathway is a key molecular mechanism of inhibiting tissue regenerative processes specifically activated in microgravity, we sought to fly a CDKN1a/p21 null mice in microgravity to test the hypothesis that this gene is required for the arrest of regenerative bone formation observed in microgravity. Specifically, we hypothesized that CDKN1a/p21 is the key effector of cell cycle arrest in bone tissue response to the stresses of spaceflight in microgravity and more broadly that CDKN1a/p21 is required for bone regenerative arrest in mice during spaceflight. To test this hypothesis, we proposed the following specific goals: 1) To determine the extent of osteoblasts proliferation, differentiation, and mineral deposition in microgravity compared to 1g in a CDKN1a/p21-null background versus WT 2) To determine if the coupling and balance of increased osteoclastogenesis and decreased osteoblastogenesis in microgravity is altered in a CDKN1a/p21-null background 3) To determine if CDKN1a/p21 expression in microgravity is regulated via mRNA stabilization by the microgravity-induced Cold Induced RNA Binding Protein (CIRBP) in response to oxidative stress in bone. The experimental approaches originally proposed in 2014 to achieve these aims include MicroComputed tomography, bone histology, and histomorphometry of injected Calcein plus BromodeoxyUridine labels to determine bone growth, and bulk bone tissue genomics. We have however also recently updated our analysis plans to include novel methodologies that now allow us to quantify at the single cell level gene expression in the bone marrow stem cell niche containing various stages of differentiation of the osteoprogenitor and hematopoietic lineages. The combined results of these studies will allow us to definitively determine if the presence of CDKN1a/p21 during spaceflight induces arrest of bone growth and regeneration mediated by CDKN1a/p21 inhibition of osteoblastic proliferation and differentiation, and if so, how this might be coupled to osteoclastic activity induced in space. In addition, for the first time, these studies seek to connect the potential effector molecular mechanism of regenerative arrest in bone (by CDKN1a/p21) to its inducer, hypothesized here to be oxidative (and other) stress response gene CIRBP stabilization of CDKN1a/p21 message RNA. When completed, these studies will provide molecular mechanistic evidence for how CDKN1a/p21, one of the most important master regulatory genes of stem cell regenerative health, is modulated by microgravity and space radiation, and could be the basis for development of preventive treatments designed to counter tissue regenerative deficits in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Recent advances in the scientific understanding of stem cells and their role in tissue regeneration created a new field of regenerative medicine that holds great promise both in the treatment, as well as prevention, of many degenerative diseases often associated with aging. In short duration human spaceflight, when the body is weightless and unloaded in microgravity, multiple tissue degenerative conditions occur, such as muscle atrophy, bone loss, immune deficiencies, and others that may have a deficit in stem cell-based tissue regeneration in common. Likewise, on Earth, when we increase mechanical loading of the body with exercise we observe an increase in adult stem cell-based tissue regeneration, such as in bone strengthening, muscle mass increase, and increased production of immune cells and red blood cells from hemotopoeietic stem cell progenitors. In total, spaceflight and ground experimental evidence from human and animal studies now strongly suggests that stem cell-based regenerative health is regulated by mechanical loading of the body, increasing with exercise and load, and decreasing with inactivity and unloading, as in microgravity. Our proposed experimental work in microgravity flight experiments and mechanical unloading models addresses the molecular, cellular, and tissue aspects of the hypothesis that gravity mechanical loading regulates regenerative health. The novel results we obtained identify the transition between stem cell progenitors and differentiated cells as a likely cellular process regulated by mechanical loading and specific molecular candidates such as the p21/CDKN1a pathway induced in microgravity as inhibitory of stem cell-based tissue regeneration in bone. This work is highly significant to NASA space exploration and medicine on Earth because it holds the promise of greater scientific understanding of how to maintain and monitor health during future voyages of space exploration, and solidly establishes Earth gravity as a major shaper of human life through promoting tissue regenerative health.
	SPACEFLIGHT AND ASSOCIATED GROUND CONTROL EXPERIMENTS: The 2020-2021 year was an extremely productive period for this grant as we finally were able to successfully conduct the proposed Cdkn1a-null mouse spaceflight experiment on the SPX-21 flight to the International Space Station (ISS) between December 2020 and January 2021. This success followed a difficult 2019-2020 year limited by COVID access restrictions to NASA facilities, until we obtained a June 2020 "RR10 Return to Work Plan" approval. Starting in October 2020, we were able to conduct all the necessary preflight tests, plus preparation of Principal Investigator (PI) provided reagents and logistical operations in time for launch. Because the first SPX-21 launch date planned for October was postponed, the transgenic mice in the first of four cohorts could not be used for flight and were quickly added to our NASA Ames Research Center (ARC) ground Institutional Animal Care and Use Committee (IACUC) protocol (animal study protocol), and were redirected to be used for a supplemental ground control of simulated microgravity using the hindlimb unloading model at the NASA ARC Animal Care Facility (ACF) in November 2020. Although we had conducted a previous large-scale ground-science hindlimb unloading experiment in 2016-2017, while the RR-10 flight opportunity was on hold, it had not included the single cell genomics analyses that we can now readily perform (nor was it conducted with the same cohort of mice bred for the RR-10 flight, or the same schedule and specific Calcein, Alizarin, and BrdU reagents for injections). This experiment was designed to allow us to compare how the hindlimb unloading model on Earth differs in bone effects caused by spaceflight in real microgravity in the presence of space radiation, albeit at low levels in low Earth orbit. In December 2020, as the RR-10 spaceflight became imminent, we

Task Progress:

deployed a team at Roskamp Institute in Sarasota Florida, where the ground control for the flight was conducted using NASA rodent habitats inside environmental chambers, and another team at NASA Ames where we conducted the vivarium control, (differing from the Rodent Habitat ground control in the use of normal vivarium cages and standard diet instead of the NASA food bars). This control was intended to account for any differences in bone parameters that may be due to the significant differences between the NASA rodent habitat and standard vivarium cage housing. With the experiment duration set at 30 days, both teams in California and Florida started acclimation and injections 2 weeks ahead of "launch", then conducted basal control group dissections around the time of launch, plus additional injections at day 30 minus 7 days and day 30 minus 2 days, and dissections again at day 30. Mice were also injected and dissected on orbit by astronauts with hindlimbs either fixed in RNALater or Formaldehyde and with the femur shaft segment cut out and centrifuged on ISS to release bone marrow cells into cryopreservation medium (serum:DMSO) to allow single cell genomic analyses back on Earth. The RR-10 ISS experiment was groundbreaking because it demonstrated, for the first time, the technical ability to collect and live-cryopreserve dissected mouse bone marrow tissue in space for single cell RNAseq genomic analyses back on Earth.

Having utilized only mice from cohort 2 for flight, and with cohorts 3 and 4 now available, and with NASA Space Biology concurrence, we quickly submitted additional IACUC protocol amendments to allow us to conduct exercise experiments in parallel to the flight disuse experiment on ISS, thus collecting a full set of bone-centric samples relevant to the study of how physical activity and disuse, plus radiation in space, regulate bone regenerative dynamics and physiology. Specifically, we conducted an additional running wheel experiment at the NASA Ames ACF and a treadmill experiment at Rensselaer Polytechnic Institute. When we concluded most animal work around February 2021, we now had samples for both WT and Cdkn1a-null: Basal Controls; Hindlimb Unloading; Normally loaded HU cage control; Spaceflight in Rodent Habitat; Ground Control in Rodent Habitat; Vivarium control; Running Wheel; Static Wheel; and Treadmill. A final set of remaining WT, and Cdkn1a-null mice from the flight cohorts was aged for a year until September 2021 and then dissected to provide hindlimb and marrow samples that may serve as a reference for correlating effects of spaceflight to aging on Earth. The various bone and marrow tissue samples we obtained from nearly 200 experimental mice is the most comprehensive set ever obtained under similar controlled conditions to investigate how exercise and disuse on Earth compare to disuse and potentially radiation exposure on the ISS spaceflight environment of LEO, and how they all correlate to aging. Although many of these potential studies are out of scope for our existing grant, we felt it was important to take advantage of the opportunity to utilize these mice and preserve these samples as part of an extensive comparable set. Finally, in a further effort to maximize the utility of the mouse cohorts bred for the RR-10 flight that were not used, we also established a collaboration with the Jackson Lab and were able to generate various mattings and mouse embryonic stem cell lines for Cdkn1a null mice, as well as matched WT controls, both male and female. These stem cell lines are intended to develop a future line of in-vitro experiments with organoids to evaluate the long-term effects of deep space radiation on tissue regeneration and the role that Cdkn1a may play in that process, this time also addressing sex as a potential factor that was not logistically possible to study with the current flight experiment and flight habitat limitations.

RR-10 SAMPLE PROCESSING and ANALYSIS PROGRESS:

MicroComputed Tomography of Bone – Starting in February 2021, once spaceflight samples had been returned to Earth on SPX-21 and received at NASA ARC, we started the extensive task of X-ray micro-computed tomography, scanning all experimental samples to evaluate bone structural parameters following spaceflight and various other related controls. Specifically, between February and September 2021 we completed a full set of 4-micron (medium resolution) scans from 200 or so complete mouse hindlimbs. The composite "oversize" hindlimb scans include the complete pelvis, femur, tibia-fibula, and foot, as well as the knee and hip joints and required about 2000 hours of X-ray beam time to complete. In addition to scanning, we have also performed tomographic reconstructions of the 4-micron scans. During the 2020-2021 period, we also initiated analysis of bone mineral parameters for cancellous bone in the distal femur, and cortical bone in the femur midshaft, although this work is still in early stages. The RR-10 hindlimb CT scans and reconstructions we obtained are unique in their completeness and for including, at relatively high-resolution, wide view of the bone position relationships to muscle and joints under multiple spaceflight and related loading conditions. Preliminary results from a limited femur sample set appears to show the expected cancellous bone loss during spaceflight, but more interesting, little or no change in Cdkn1a null mice, suggesting deletion of the gene may have rescued sufficient bone formation by osteoblasts to counteract bone loss in space. Additional CT scans of the proximal and distal femur at 1.5-micron high-resolution are currently being performed which will provide structural fine detail of lacunar size, number, and distribution, as well as osteoclast bone resorption sites in the bone surface. Bone and Bone Marrow Genomic Analyses – In the current reporting period, genomic analysis of RR-10 samples was started in June 2021 (as logistical and supply chain procurement obstacles during COVID delayed our ability to procure the necessary reagents, supplies, and kits for molecular biology benchwork). From June to September 2021, we completed pilot mRNA isolations from bulk dissected clean pelvic bone from the Ileum region and from the cryopreserved bone marrow cells for 10X Genomics single cell mRNAseq analyses. Both pilot experiments spanned a main set of 16 samples each of the different genotypes and principal loading conditions and were transported to Illumina sequencing libraries and quality controlled with limited sequencing on an Illumina MySeq instrument. After pilot sequencing library quality control evaluations were successfully conducted, validating our methods, we resumed library generation for upcoming sequencing by the NASA GeneLab team using NovaSeq instrument and S4 flow cells. Single cell and bulk tissue sequencing by GeneLab are expected to be completed by the end of February 2022. The on-orbit isolation and high-viability, live, cryopreservation of bone marrow stem cells from the shaft of mouse femurs was a major technical spaceflight accomplishment that will allow the investigator team to study gene expression as it was in space, not subjected to the stresses of reentry, splashdown, and lengthy live animal return to ground-based laboratories.

Bone Dynamic Histomorphometry – Starting in June 2021, as we prepared the first set of 16 single cell RNAseq Illumina libraries, the femoral shafts containing the bone marrow used for single cell cryopreservation were recovered, and separately fixed, and dehydrated in an ethanol series before being transferred to a clearing agent (methyl salicylate) for confocal imaging. Following tissue clearing, the calcein injection fluorescent label in intact semi-transparent bone was readily imaged in the confocal microscope as separate lines, indicating the quantitative extent of bone growth during spaceflight and ground control conditions. Currently, at the end of the reporting period, only two of ten 18-sample sets have been processed, but preliminary limited results suggest arrest of bone formation in WT mice, but not in Cdkn1a null mice, potentially supporting the main grant hypothesis that Cdkn1a is a key mediator of bone formation arrest in space.

Histology of Bone Tissue – We expect to be able to start processing samples for histology in January 2022.

Public Outreach by Open Science – Tomographic reconstruction datasets will be made available to the public for open

science use via the NASA Ames Life Sciences Data Archive (ALSDA) / GeneLab platform. Genomic datasets will also be made available later in the project.

Collaboration/PI-Led RR-10 Biospecimen Sharing Dissections – To increase the utility to science of the mice used for RR-10, the PI team and NASA Space Biology organized a Biospecimen Sharing Program (BSP) at the NASA Ames PI lab in April 2021. The PI-selected team of collaborators included Dr. Grace O'Connel and Shiyin Lim from University of California Berkeley (vertebrae, vertebral discs, and femoral head cartilage); Dr. Mike Pecaut from Loma Linda University (Thymus, Spleen); Dr. Xiao (Vivian) Mao from Loma Linda University (R. Eye); Dr. Eleonora Grigoryan from the Koltsov Institute of Developmental Biology (Moscow) of the Russian Academy of Sciences (L. Eye and Muscle); Dr Karen Jonscher at University of Oklahoma Health Sciences Center (Liver); Dr. Egle Cekanaviciute at NASA ARC (Brain); Dr. Jared Broderick at NASA ARC (Intestine); and Dr. Elizabeth Blaber at Rensselaer Politechnic Institute (Heart). In addition to the PI-led BSP, our laboratory simultaneously supported the NASA open BSP dissection of the mouse female reproductive tract for study by Dr. Lane Christenson at University of Kansas Medical Center. Overall, the collaborations initiated or supported by the RR-10 PI team increased other investigator focus on the RR-10 flight experiment samples by a factor of 10, with results already being reported in fora such as the recent 2021 American Society for Gravitational and Space Research (ASGSR) meeting.

GROUND-BASED EXPERIMENTS:

Because the current grant was initially awarded in 2014, and because of extensive delays getting to spaceflight in late 2020, NASA Space Biology previously authorized reprogramming some of the grant funding and effort to addressing the grant specific aims using ground-based cell assays and animal experiments. Under this authority we recently initiated an effort to determine if Cdkn1a in osteoblastic (bone-forming) cells could be regulated by mechanical loading and unloading, simulating microgravity. Cdkn1a was initially discovered as an early radiation response gene that interacts with cyclins, inactivating them and arresting the cell cycle just long enough for DNA repair to occur before cells resume cell division. A subsequent knockout of the gene in 2010 revealed that it was also a strong constitutive inhibitor of tissue regeneration, with the transgenic mice being nearly normal but with remarkable wound healing and closure abilities, as well as limb extremity regrowth capacity, akin to regenerative amphibians. When our laboratory discovered Cdkn1a was elevated several-fold in bone (and heart) following spaceflight on STS-131 in 2010, we hypothesized that the gene might be activated in space to arrest the cell cycle in regenerative osteoblasts in response to either space radiation or microgravity unloading or a synergistic combination of both spaceflight factors. While the RR-10 spaceflight experiment to test this hypothesis awaited a flight opportunity, we developed an experiment concept to determine if mechanical loading and unloading could regulate Cdkn1a expression like radiation does. We developed and fabricated computer controlled cyclic stretch cell culture devices that simulated the deformation of the semi-rigid bone surface associated with walking, running, jumping, stretching, and contracting by 0.01% at a rate of 1 Hz. We then plated primary bone marrow stromal osteoprogenitor cells from WT and Cdkn1a null mice in the devices and conducted 21-day mineralization assays induced by ascorbic acid and beta glycerol phosphate treatment. In total we studied proliferation, differentiation, mineralization and stemness of the cells, both at the population level with traditional cell biology methods and using single cell RNAseq analyses. Using this approach, we were able to specifically study single cell sub-populations that fit the gene biomarker pattern for osteoprogenitors, early osteoblasts, and late mineralizing osteoblasts, both in WT and Cdkn1a null cells, with and without cyclic stretch. Remarkably, we discovered that in addition to being elevated in response to radiation, Cdkn1a expression is suppressed by stretch mechanical loading, and elevated under static culture conditions, analogous to microgravity. Furthermore, in the absence of Cdkn1a, mechanical stimulation results in a greatly enhanced osteogenesis and mineralization. These seminal findings were published in the Journal, "Stem Cell Research" at the end of the reporting period in August 2021 [Ed. Note: see Bibliography]. These results also mean that Cdkn1a has the potential to act as the master negative key cell cycle molecular mechanism regulating bone regenerative cell proliferation in direct response to the two main spaceflight factors, space radiation and microgravity unloading, potentially opening the door to gene-specific pharmacological inhibitor countermeasures. Conversely, this finding also means that physical exercise can release cell proliferation arrest and contribute to bone formation via stimulation of osteogenesis. Finally, because of this ground-based work our laboratory developed the technical ability to conduct and analyze single cell RNAseq experiments, as well as the genomic marker sets for isolating specific osteogenesis stages in the bone marrow mesenchymal cell population separately from all other cell lineages present, such as hematopoietic progenitors that can strongly bias bulk tissue genomic analyses. That single cell capability is now being utilized to analyze the bone marrow samples collected on orbit during RR-10.

Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Almeida EAC, Juran CM. "In-vitro cyclic stretch mechanical loading regulates p21/CDKN1a expression and differentiation progression of regenerating osteogenic lineages but not early mesenchymal osteoprogenitors." 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. ID 171. , Nov-2020
Articles in Other Journals or Periodicals	Almeida EAC, Juran CM. "Stem cell-based tissue regenerative health in space." National Academies of Sciences, Engineering, and Medicine. To appear in special decadal survey on biological and physical sciences research in space 2023-2032. In review. , Jun-2023
Articles in Peer-reviewed Journals	Juran CM, Zvirblyte J, Cheng-Campbell M, Blaber EA, Almeida EAC. "Cdkn1a deletion or suppression by cyclic stretch enhance the osteogenic potential of bone marrow mesenchymal stem cell-derived cultures." Stem Cell Res. 2021 Oct;56:102513. https://doi.org/10.1016/j.scr.2021.102513 ; PMID: 34517335 , Oct-2021
Articles in Peer-reviewed Journals	Kumar A, Tahimic CGT, Almeida EAC, Globus RK. "Spaceflight modulates the expression of key oxidative stress and cell cycle related genes in heart." Int J Mol Sci. 2021 Aug 23;22(16):9088. https://doi.org/10.3390/ijms22169088 ; PMID: 34445793; PMCID: PMC8396460 , Aug-2021